

Computational Sociology

Fundamentals of natural language processing

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Course updates

Homework

- ▶ Homework 1 due today
 - ▶ Submit by committing and pushing to your GitHub homework repository
 - ▶ Check all required components added

Course updates

Project proposal

- ▶ 2-3 page double-spaced proposal for final project
 - ▶ Introduction and related work
 - ▶ Research question
 - ▶ Data
 - ▶ Methodology
 - ▶ References
- ▶ Projects must involve at least one methodology covered during the class
 - ▶ Ideally collect new data and conduct a computational analysis
- ▶ Submit PDF via email, due 10/2

Plan

- ▶ Part I: What is natural language processing?
- ▶ Part II: Working with text
- ▶ Part III: Comparing documents

Part I: What is NLP?

Introduction to NLP

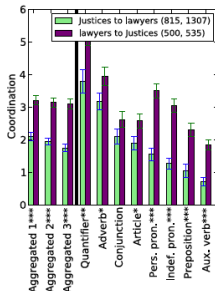
What is natural language processing?

- ▶ Three components of NLP*:
 - ▶ Natural language / “text as data”
 - ▶ A corpus of text (e.g. books, reviews, tweets, e-mails)
 - ▶ (Computational) linguistics
 - ▶ Linguistic theory to guide analysis and computational approaches to handle data
 - ▶ Statistics
 - ▶ Statistical approaches to text data

*Not *that* NLP: https://en.wikipedia.org/wiki/Neuro-linguistic_programming

Introduction to NLP

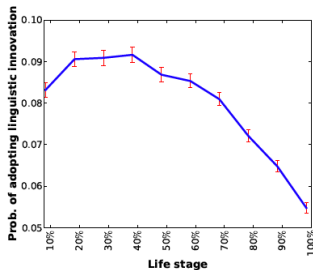
Applications: Power dynamics



Danescu-Niculescu-Mizil, Cristian, Lillian Lee, Bo Pang, and Jon Kleinberg. 2012. "Echoes of Power: Language Effects and Power Differences in Social Interaction." In *Proceedings of the 21st International Conference on World Wide Web*, 699–708.

Introduction to NLP

Applications: Identity and group membership



(c) Adoption of lexical innovations

Danescu-Niculescu-Mizil, Cristian, Robert West, Dan Jurafsky, Jure Leskovec, and Christopher Potts. 2013. "No Country for Old Members: User Lifecycle and Linguistic Change in Online Communities." In *Proceedings of the 22nd International Conference on World Wide Web*, 307–18.

Introduction to NLP

Applications: Trust and betrayal

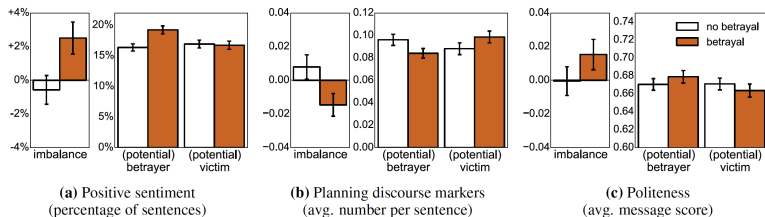
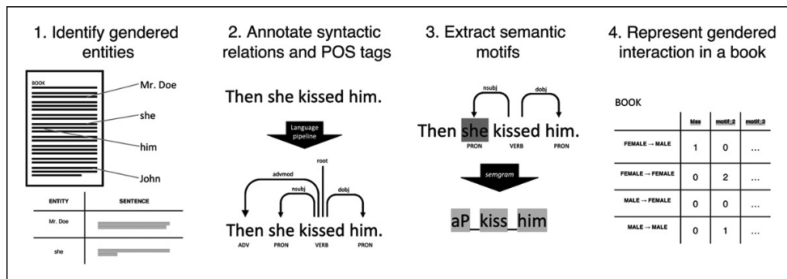


Figure 3: Friendships that will end in betrayal are imbalanced. The eventual betrayer is more positive, more polite, but plans less than the victim. The white bars correspond to matched lasting friendships, where the roles of potential betrayer and victim are arbitrarily assigned; in these cases, the imbalances disappear. Error bars mark bootstrapped standard errors (Efron, 1979).

Niculae, Vlad, Srijan Kumar, Jordan Boyd-Graber, and Cristian Danescu-Niculescu-Mizil. 2015. "Linguistic Harbingers of Betrayal: A Case Study on an Online Strategy Game." In *Proceedings of the 53rd Annual Meeting of the ACL and the 7th International Joint Conference on Natural Language Processing*.

Introduction to NLP

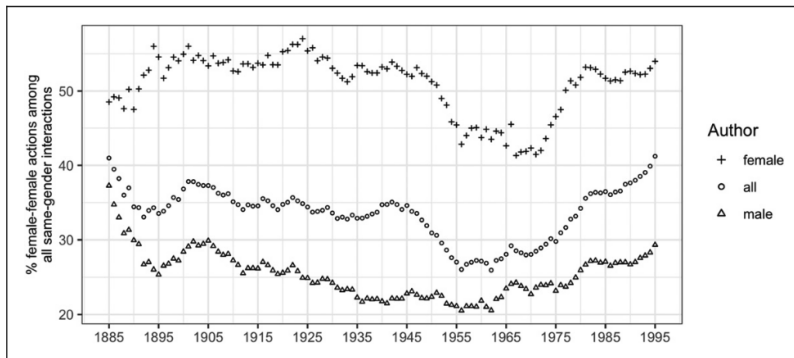
Applications: Gender and social action



Stuhler, Oscar. 2022. "Who Does What to Whom? Making Text Parsers Work for Sociological Inquiry." *Sociological Methods & Research* 51 (4): 1580–1633.

Introduction to NLP

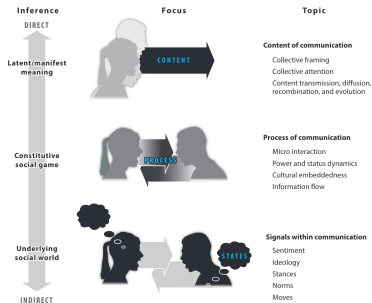
Applications: Gender and social action



Stuhler, Oscar. 2022. "Who Does What to Whom? Making Text Parsers Work for Sociological Inquiry." *Sociological Methods & Research* 51 (4): 1580–1633.

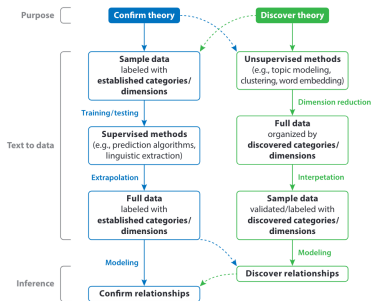
Introduction to NLP

NLP and Social Theory (Evans and Aceves 2016)



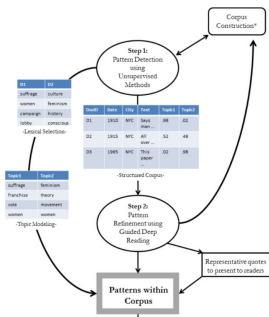
Introduction to NLP

NLP and Social Theory (Evans and Aceves 2016)



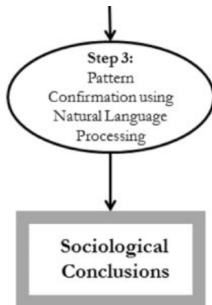
Introduction to NLP

Computational Grounded Theory (Nelson 2020)



Introduction to NLP

Computational Grounded Theory (Nelson 2020)



Introduction to NLP

Computational and qualitative research

Computational approaches are sometimes less subtle and deep than the reading of a skillful analyst, who interprets text in context. Nevertheless, . . . recent advances in NLP and ML are being used to enhance qualitative analysis in two ways. First, supervised ML prediction tools can “learn” and reliably extend many sociologically interesting textual classifications to massive text samples far beyond human capacity to read, curate, and code. Second, unsupervised ML approaches can “discover” unnoticed, surprising regularities in these massive samples of text that may merit sociological consideration and theorization.

James Evans and Pedro Aceves, 2016

Introduction to NLP

Text as data

Table 1 Four principles of quantitative text analysis

-
- (1) All quantitative models of language are wrong—but some are useful.
 - (2) Quantitative methods for text amplify resources and augment humans.
 - (3) There is no globally best method for automated text analysis.
 - (4) Validate, Validate, Validate.
-

Justin Grimmer and Brandon Stewart, 2013

Introduction to NLP

NLP class schedule

- ▶ Week 5 (today)
 - ▶ Pre-processing, bag-of-words, and the vector-space model
- ▶ Week 6
 - ▶ Word embeddings
- ▶ Week 7
 - ▶ Topic models
- ▶ Week 9 (after fall break)
 - ▶ Supervised text classification
- ▶ Weeks 10-12
 - ▶ Language models and multimodal models

Part II: Working with text

Working with text

Pre-processing

- ▶ There are several steps we need to take to clean or “pre-process” texts for analysis
 - ▶ Tokenization
 - ▶ Stemming/lemmatization
 - ▶ Stop-word removal
 - ▶ Handling punctuation and special characters

Working with text

Tokenization

- ▶ Tokenization is the process of splitting a document into words
 - ▶ e.g. “Cogito, ergo sum” $\Rightarrow$ (“Cogito,” , “ergo” , “sum”)
- ▶ Splitting on whitespace is a simple starting point in English
 - ▶ Tokenizers also need rules for punctuation, contractions, and hyphenated words
- ▶ Tokenization is more difficult in languages like Mandarin where word boundaries need to be detected

Working with text

Stemming/lemmatization

- ▶ We often want to reduce sparsity by reducing words to a common root
 - ▶ e.g. (“school”, “schools”, “schooling”, “schooled”) $\Rightarrow$ “school”
- ▶ Stemming is a simple, heuristic-based approach
- ▶ Lemmatization is a more rigorous approach based on morphology, but computationally-intensive

Working with text

Stop-word removal

- ▶ “Stop-words” are frequently occurring words that are often removed
- ▶ The intuition is that they add little meaning and do not help us to distinguish between documents
 - ▶ e.g. Virtually all texts in English will contain the words “and”, “the”, “of”, etc.
- ▶ Most NLP packages have lists of stop-words to easily facilitate removal

Working with text

Handling punctuation and special characters

- ▶ In many cases we may want to remove punctuation, HTML tags, or unwanted control characters
 - ▶ This is often done using regular expressions (“regex”)
 - ▶ Words are typically set to lowercase

Working with text

Pre-process with caution!

- ▶ Researchers often apply these techniques before starting an analysis, but these choices may affect our results
 - ▶ There is no one-size-fits-all solution, so think carefully before removing anything
 - ▶ It's often useful to experiment to see if pre-processing steps affect results

Working with text

Pre-process with caution!

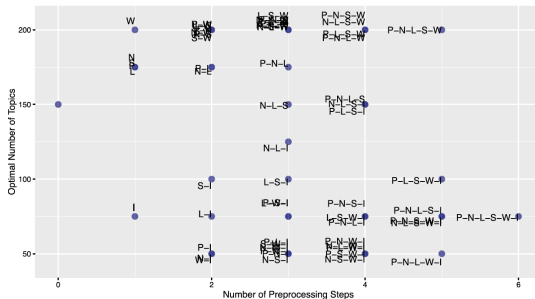


Figure 2. Plot depicting the optimal number of topics (as selected via perplexity) for each of 64 preprocessing specifications not including trigrams. On the x-axis is the number of preprocessing steps, and the y-axis is the number of topics. Each point is labeled according to its specification.

Denny, Matthew J., and Arthur Spirling. 2018. "Text Preprocessing For Unsupervised Learning" *Political Analysis* 26 (02): 168–89. <https://doi.org/10.1017/pan.2017.44>.

Working with text

Word counts

- ▶ Now we have done some pre-processing, one of the most basic ways we can start to analyze texts is by counting the frequency of words.
 - ▶ e.g. “I think, therefore I am” $\Rightarrow$

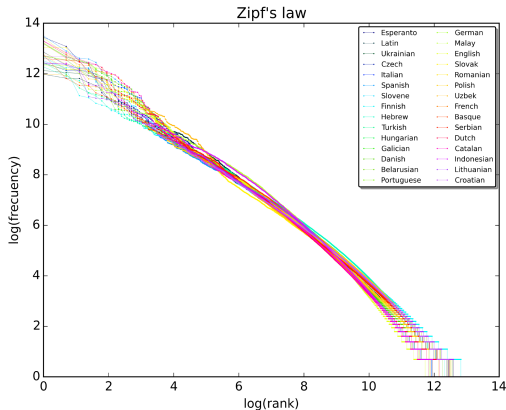
Word	Count
I	2
think	1
therefore	1
am	1

Working with text

Frequency distributions

- ▶ **Zipf's law:** *A word's frequency is inversely proportional to its rank order in the frequency distribution.*
 - ▶ “the” is the most common word in the English language, accounting for 7% of all words in the *Brown Corpus of American English*
 - ▶ “and” and “of” compete for second place, each accounting for ~3.5% of words in the corpus
 - ▶ The most frequent 135 words account for approximately half the 1 million words in the corpus
 - ▶ Around 50,000 words, representing half the total unique words in the corpus, are *hapax legomena*, words which only occur once

Zipf's law



A plot of the rank versus frequency for the first 10 million words in 30 Wikipedias (dumps from October 2015) in a log-log scale (Source: Wikipedia).

Working with text

Bag-of-words

- ▶ Documents are often treated as “bags of words”, i.e. a collection of words without information about the order
 - ▶ e.g. “This is a document” $\Rightarrow$ (“a”, “document”, “is”, “This”, “is”)

Working with text

Example: Loading data

```
library(tidyverse)
library(tidytext)
library(gutenbergr)
library(ggplot2)
library(stringr)
#install.packages("tm") # Dependency for tidytext, required for cast_dt

ef <- gutenberg_download(41360) # Download Elementary Forms
cm <- gutenberg_download(61) # Download Communist Manifesto

ef$title <- "Elementary Forms"
cm$title <- "Communist Manifesto"

texts <- bind_rows(ef, cm)
```

Working with text

We are using a tabular format where each text is represented as a table, where the first column is the ID in the Project Gutenberg database and the text field contains each line as a new row.

```
tail(texts$text)
```

```
## [1] "declare that their ends can be attained only by the forcible ov  
## [2] "of all existing social conditions. Let the ruling classes tremb  
## [3] "Communitic revolution. The proletarians have nothing to lose b  
## [4] "chains. They have a world to win."  
## [5] ""  
## [6] "WORKING MEN OF ALL COUNTRIES, UNITE!"
```


Working with text

Tokenizing using tidytext

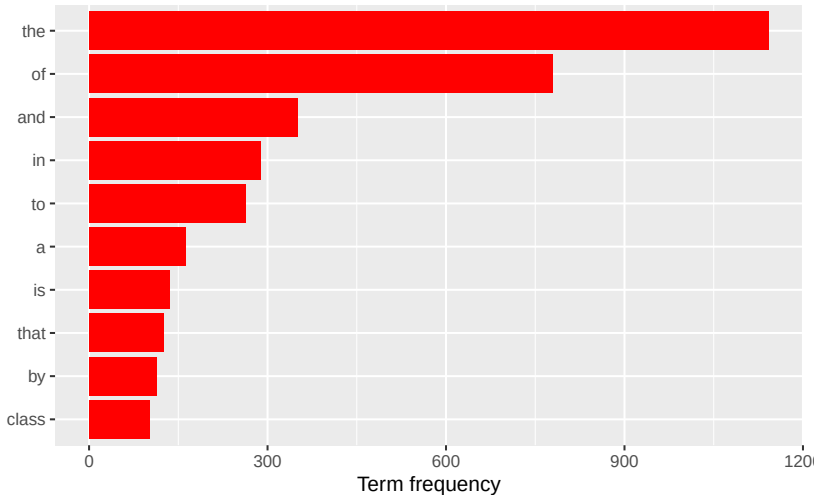
The `unnest_tokens` function is used to tokenize the text, resulting in a table containing the original text ID and each token as a separate row.

```
tidy.text <- texts |> unnest_tokens(word, text)
tail(tidy.text$word)
```

```
## [1] "working" "men" "of" "all" "countries" "uni
```

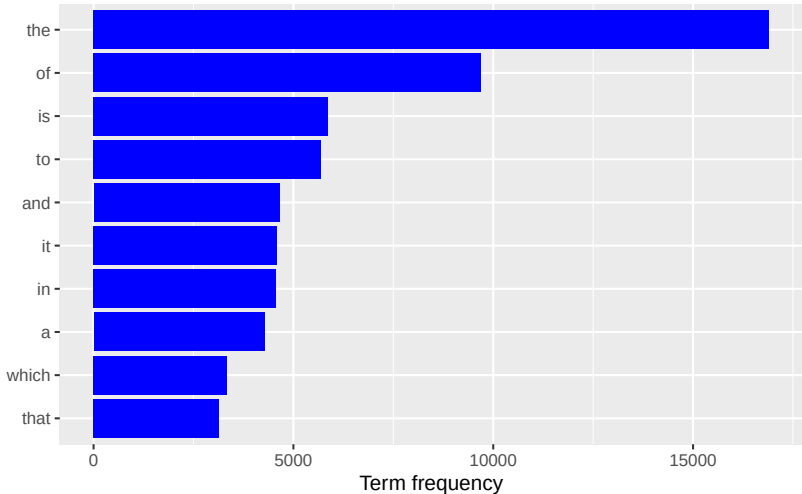
Term frequency in *The Communist Manifesto*

10 most frequent terms in The Communist Manifesto



Term frequency in *Elementary Forms*

10 most frequent terms in Elementary Forms



Working with text

Removing stopwords

We can load a corpus of stop words contained in `tidytext` and use `anti_join` to filter our texts. This retains all records without a match in stopwords.

```
data(stop_words)
head(stop_words)
```

```
## # A tibble: 6 x 2
##   word      lexicon
##   <chr>    <chr>
## 1 a       SMART
## 2 a's     SMART
## 3 able    SMART
## 4 about   SMART
## 5 above   SMART
## 6 according SMART
```

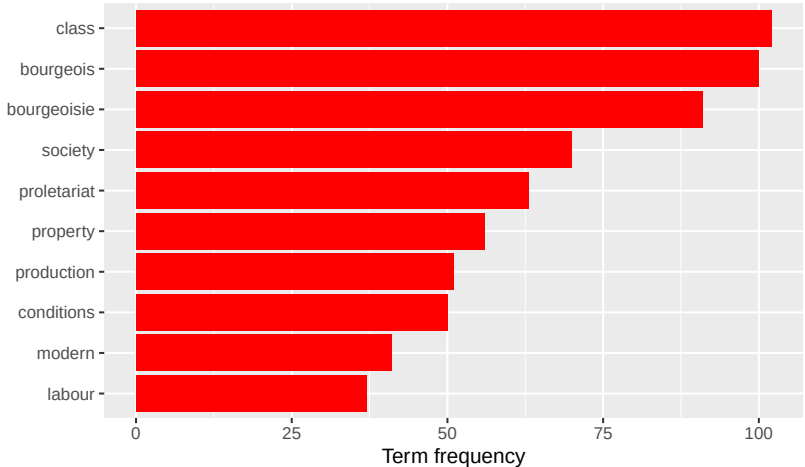
Working with text

Removing stopwords

```
tidy.text <- tidy.text |>  
  anti_join(stop_words)
```

Term frequency in *The Communist Manifesto*

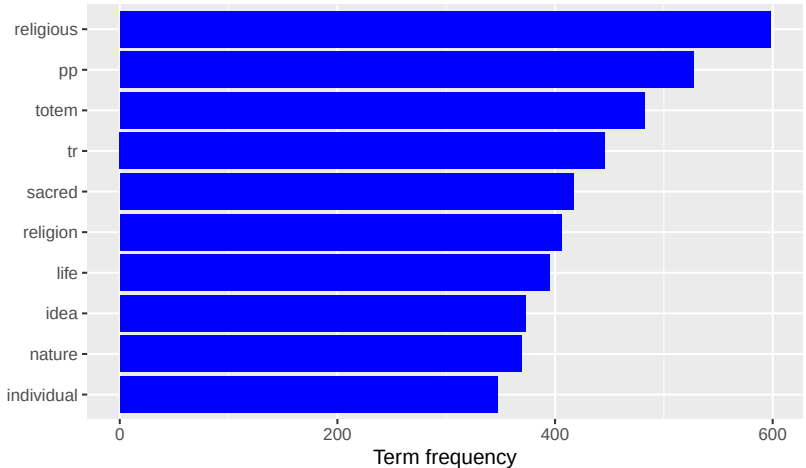
10 most frequent terms in The Communist Manifesto



Stopwords removed

Term frequency in *Elementary Forms*

10 most frequent terms in Elementary Forms



Stopwords removed

Working with text

Removing new stopwords

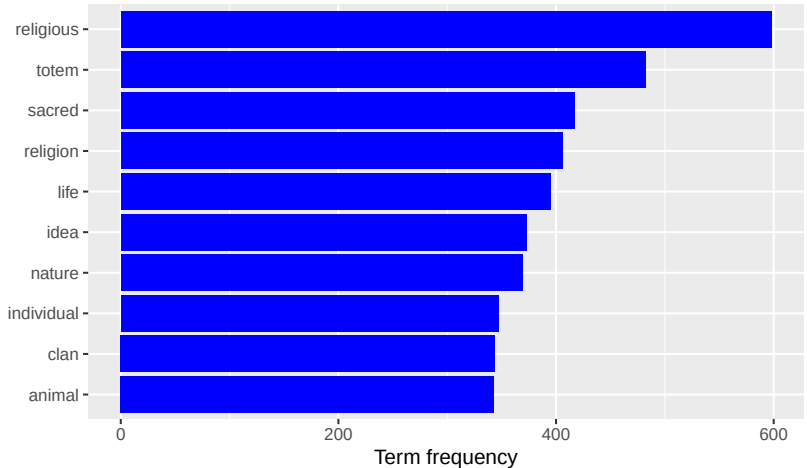
The last example shows how there is still some “junk” in the Durkheim text. We can try to remove this by adding to our stopwords list.

```
to.remove <- tibble(text=c("pp", "tr")) |> unnest_tokens(word, text)

tidy.text <- tidy.text |>
  anti_join(to.remove)
```


Term frequency in *Elementary Forms*

10 most frequent terms in Elementary Forms



Stopwords removed+

Working with text

Stemming

We can stem the terms using a function from the package `SnowballC`, which is a wrapper for a commonly used stemmer called the Porter Stemmer, written in C.

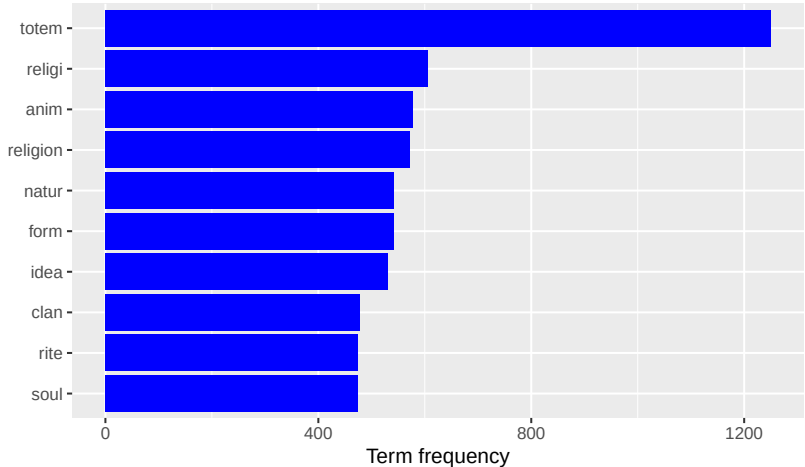
```
library(SnowballC)

tidy.text <- tidy.text |>
  mutate(word = wordStem(word, language = "en"))
```

Stemmer example: https://cbail.github.io/SICSS_Basic_Text_Analysis.html. More on stemming and lemmatizing in R: <https://smltar.com/stemming>.

Term frequency in *Elementary Forms*

10 most frequent terms in Elementary Forms



Stopwords removed+, stemmed

Working with text

Counting words

Let's get counts of words across both texts to analyze their distribution.

```
# Count words by text
text.words <- tidy.text |> count(title, word, sort = TRUE)

# Get total number of words in each text
total.words <- text.words |> group_by(title) |>
  summarize(total = sum(n))

# Merge
words <- left_join(text.words, total.words)
```

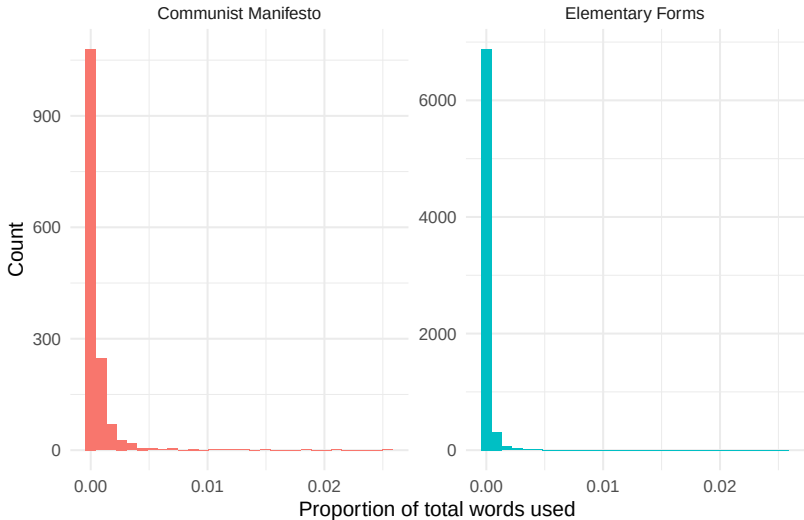
Working with text

Counting words

```
head(words)
```

```
## # A tibble: 6 x 4
##   title          word      n total
##   <chr>         <chr>  <int> <int>
## 1 Elementary Forms totem    1250 78807
## 2 Elementary Forms religi    606 78807
## 3 Elementary Forms anim     577 78807
## 4 Elementary Forms religion  572 78807
## 5 Elementary Forms form     542 78807
## 6 Elementary Forms natur     542 78807
```

Word frequency distribution



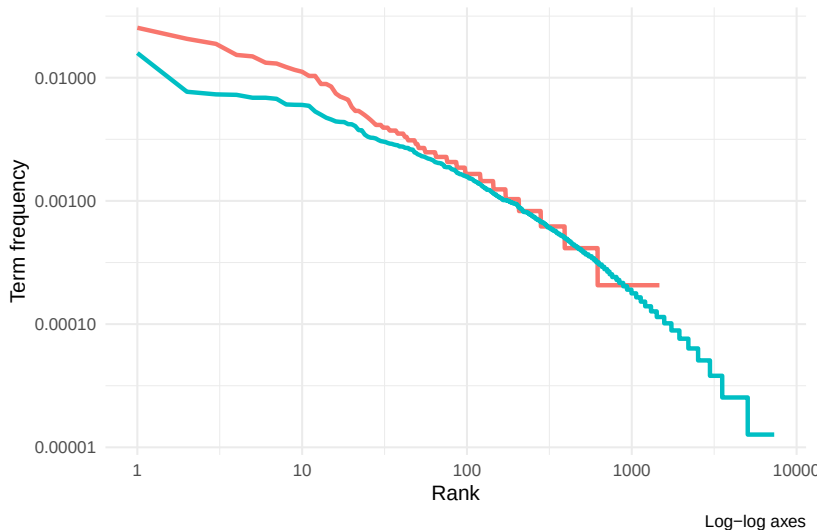
Working with text

Zipf's law

Calculating rank and frequency for each word in each text.

```
freq_by_rank <- words |>  
  group_by(title) |>  
  mutate(rank = row_number(),  
         `term frequency` = n/total) |>  
  ungroup()
```

Zipf's law



Working with text

N-grams

- ▶ So far we have just considered treating a text as a “bag-of-words”
- ▶ One way to maintain some information about word order (and hence syntax) is to use N-grams
- ▶ An *N-gram** is a sequence of N words
- ▶ We often split texts into N-grams to capture basic syntactic units like phrases
 - ▶ N is usually small.
 - ▶ $N = 2$ is called a “bigram”; $N = 3$ is a “trigram”
 - ▶ e.g. “I like peanut butter” contains the following bigrams: “I like”, “like peanut”, & “peanut butter”.

*Nothing to do with Scientology [https://en.wikipedia.org/wiki/Engram_\(Dianetics\)](https://en.wikipedia.org/wiki/Engram_(Dianetics))

Working with text

N-grams

- ▶ We can also use *character N-grams* to split documents into sequences of characters
 - ▶ e.g. “character” can be split into the following triplets (“cha”, “har”, “ara”, “rac”, “act”, “cte”, “ter”)
- ▶ BERT uses WordPiece tokenization with a learned vocabulary of words and subword pieces
 - ▶ An unfamiliar word can often be split into familiar pieces
 - ▶ Words that cannot be represented receive an unknown token ([UNK])

Exercise

Modify `unnest_tokens` to obtain trigrams from the texts. Use `?function_name` for help.

Trigrams in *The Communist Manifesto*

```
tidy.trigrams |> filter(gutenberg_id == 61) |>  
  count(word, sort = TRUE) |>  
  slice(1:10) |>  
  mutate(word = reorder(word, n)) |>  
  ggplot(aes(n, word)) +  
  geom_col(fill='red') +  
  labs(y = NULL, x='Term frequency',  
       title="Top 10 trigrams in The Communist Manifesto")
```

Trigrams in *Elementary Forms*

```
tidy.trigrams |> filter(gutenberg_id == 41360) |>  
  count(word, sort = TRUE) |>  
  slice(1:10) |>  
  mutate(word = reorder(word, n)) |>  
  ggplot(aes(n, word)) +  
  geom_col(fill='blue') +  
  labs(y = NULL, x='Term frequency',  
       title="Top 10 trigrams in Elementary Forms")
```

Part III: Comparing documents

Working with text

Comparing documents

1. Re-weighting word counts to find distinctive words
2. Representing documents as vectors of word counts
3. Geometric interpretations of document vectors

Working with text

Limitations of word counts

- ▶ Raw word counts are an insufficient for document comparison
 - ▶ Some words occur in most documents, providing little information (recall Zipf's law)
 - ▶ Some words are rare, providing little generalizable insight
- ▶ Goal: Find words that help distinguish between documents

Working with text

Term-frequency inverse document-frequency (TF-IDF)

- ▶ Term-frequency inverse document-frequency (TF-IDF) is a way to weight word counts (“term frequencies”) to give higher weights to words that help distinguish between documents
 - ▶ Intuition: Adjust word counts to account for how many documents a word appears in

Working with text

Calculating term-frequency inverse document-frequency (TF-IDF)

- ▶ D = number of documents in the corpus
- ▶ $tf_{t,d}$ = number of times term t is used in document d , as a proportion of all words in d
- ▶ df_t = number of documents containing term t
- ▶ $idf_t = \log(\frac{D}{df_t})$
 - ▶ $\frac{D}{df_t}$, the ratio of documents to documents with t is larger for terms occurring in fewer documents
 - ▶ The logarithm penalizes very high values
 - ▶ If a word occurs in all documents
 $df_t = D \rightarrow idf_t = \log \frac{D}{D} = \log(1) = 0$.
- ▶ We then use these values to calculate $TF-IDF_{t,d} = tf_{t,d} * idf_t$

Working with text

Computing TF-IDF in tidytext

We can compute TF-IDF weights using the word-count object we created earlier.

```
tidy.tfidf <- words |> bind_tf_idf(word, title, n)
head(tidy.tfidf)
```

```
## # A tibble: 6 x 7
##   title          word      n total      tf  idf tf_idf
##   <chr>         <chr> <int> <int>   <dbl> <dbl> <dbl>
## 1 Elementary Forms totem   1250 78807 0.0159 0.693 0.0110
## 2 Elementary Forms religi    606 78807 0.00769 0      0
## 3 Elementary Forms anim     577 78807 0.00732 0      0
## 4 Elementary Forms religion  572 78807 0.00726 0      0
## 5 Elementary Forms form     542 78807 0.00688 0      0
## 6 Elementary Forms natur     542 78807 0.00688 0      0
```

Working with text

```
## # A tibble: 2 x 7
##   title                word      n total      tf    idf tf_idf
##   <chr>                <chr>  <int> <int>    <dbl> <dbl> <dbl>
## 1 Communist Manifesto countri    26  4835 0.00538      0      0
## 2 Elementary Forms    countri    16 78807 0.000203     0      0
```

Working with text

```
## # A tibble: 1 x 7
##   title          word      n total      tf    idf    tf_idf
##   <chr>         <chr>    <int> <int>    <dbl> <dbl>    <dbl>
## 1 Elementary Forms australia  108 78807 0.00137 0.693 0.000950
```

Working with text

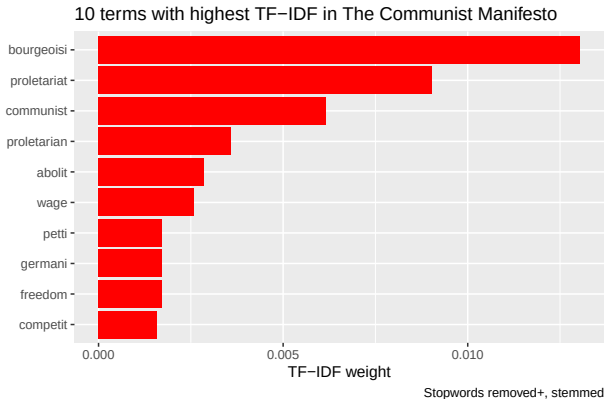
Choose another word and inspect the results.

Working with text

In this case *all* words unique to one document will have the same IDF score. Why?

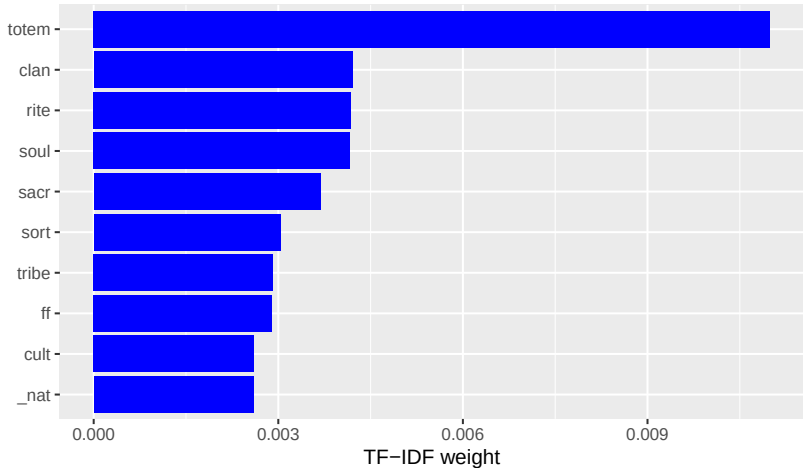
```
## # A tibble: 6 x 7
##   title          word      n total      tf    idf  tf_idf
##   <chr>         <chr> <int> <int>    <dbl> <dbl>  <dbl>
## 1 Elementary Forms totem  1250 78807 0.0159  0.693 0.0110
## 2 Elementary Forms clan   478 78807 0.00607 0.693 0.00420
## 3 Elementary Forms rite   475 78807 0.00603 0.693 0.00418
## 4 Elementary Forms soul   474 78807 0.00601 0.693 0.00417
## 5 Elementary Forms sacr   419 78807 0.00532 0.693 0.00369
## 6 Elementary Forms sort   345 78807 0.00438 0.693 0.00303
```

TF-IDF weighted word stems in *The Communist Manifesto*



TF-IDF weighted word stems in *Elementary Forms*

10 terms with highest TF-IDF in Elementary Forms



Stopwords removed+, stemmed

The vector-space model

The document-term matrix (DTM)

- ▶ A frequently used bag-of-words representation of a text corpus is the *Document-Term Matrix*:
 - ▶ Each row* is a document (a unit of text)
 - ▶ Each column is a term (word)
 - ▶ For a given DTM X , each cell $X_{i,j}$ indicates the number of times term j occurs in document i .
 - ▶ This can be the raw term counts or TF-IDF weighted counts.

*Sometimes the rows and columns are reversed, resulting in a *Term-Document Matrix* or *TDM*

The vector-space model

Casting a tidytext object into a DTM

```
X <- texts |> unnest_tokens(word, text) |>
  anti_join(stop_words) |> count(title, word) |>
  filter(n > 10) |>
  cast_dtm(title, word, n)
print(X)
```

```
## <<DocumentTermMatrix (documents: 2, terms: 1499)>>
## Non-/sparse entries: 1526/1472
## Sparsity           : 49%
## Maximal term length: 15
## Weighting          : term frequency (tf)
```

Note: This matrix is not weighted by TF-IDF, although we could apply the weights if desired.

The vector-space model

Viewing the DTM

The object created has class `DocumentTermMatrix` from the `tm` package. We can inspect this to see what it contains.

```
class(X)
```

```
## [1] "DocumentTermMatrix"      "simple_triplet_matrix"
```

```
dim(X)
```

```
## [1]      2 1499
```

The vector-space model

Viewing the DTM

The row and column names hold the documents and the vocabulary.

```
X$dimnames[1]
```

```
## $Docs
```

```
## [1] "Communist Manifesto" "Elementary Forms"
```

```
X$dimnames[[2]][1:20] # first 20 columns
```

```
## [1] "abolition" "action" "antagonisms" "bourgeois" "bourge  
## [6] "capital" "character" "class" "classes" "common  
## [11] "communism" "communist" "communists" "competition" "condit  
## [16] "countries" "country" "developed" "development" "existe
```

The vector-space model

Viewing the DTM

The easiest way to see the actual DTM is to cast it to a matrix.

```
Xm <- as.matrix(X)
```

The vector-space model

Geometric interpretations

- ▶ Each document is a vector in N -dimensional space, where N is the total number of unique words (row of the DTM / column of TDM)
- ▶ Each word is a vector in D -dimensional space, where D is the number of documents (columns of the DTM / row of TDM)

See <https://web.stanford.edu/~jurafsky/slp3/6.pdf> for more details on the vector-space model

The vector-space model

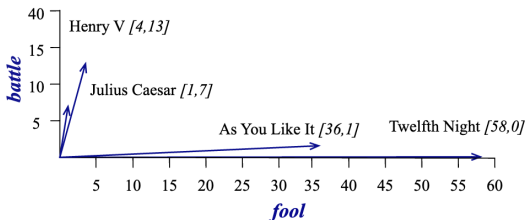
Document vectors

	As You Like It	Twelfth Night	Julius Caesar	Henry V
battle	1	0	7	13
good	114	80	62	89
fool	36	58	1	4
wit	20	15	2	3

This example from Jurafsky and Martin shows a Term-Document Matrix (TDM) pertaining to four key words from four Shakespeare plays. The document vectors are highlighted in red.

The vector-space model

Document vectors



Here vectors for each play are plotted in two-dimensional space. The y- and x-axes indicate the number of times the words “battle” and “fool” appear in each play. Note how some vectors are closer than others and how they have different lengths.

The vector-space model

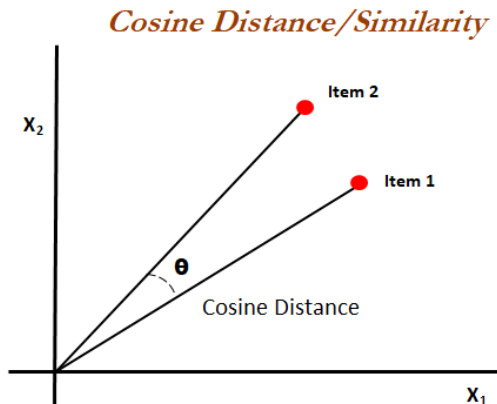
Word vectors

	As You Like It	Twelfth Night	Julius Caesar	Henry V
battle	1	0	7	13
good	114	80	62	89
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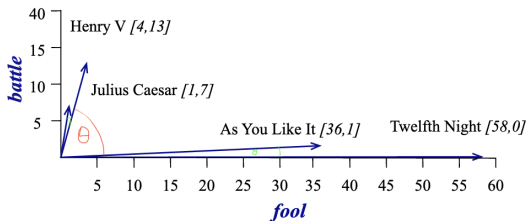
Figure 6.5 The term-document matrix for four words in four Shakespeare plays. The red boxes show that each word is represented as a row vector of length four.

We could also treat the rows of this matrix as vector representations of each word. We will return to this idea next week.

Cosine similarity



Cosine similarity



Cosine similarity

$\vec{u}$ and $\vec{v}$ are vectors representing texts (e.g. rows from a DTM matrix). We can compute the cosine of the angle between these two vectors using the following formula:

$$\cos(\theta) = \frac{\vec{u} \cdot \vec{v}}{\|\vec{u}\| \|\vec{v}\|} = \frac{\sum_i \vec{u}_i \vec{v}_i}{\sqrt{\sum_i \vec{u}_i^2} \sqrt{\sum_i \vec{v}_i^2}}$$

Cosine similarity

Calculation

```
set.seed(27516)
u <- rnorm(10)
v <- rnorm(10)

sum(u*v) / (sqrt(sum(u^2)) * sqrt(sum(v^2)))
```

```
## [1] -0.2520848
```

```
# Same result via matrix multiplication
u %*% v / (sqrt(u %*% u) * sqrt(v %*% v))
```

```
##           [,1]
## [1,] -0.2520848
```

Cosine similarity

Making a function

```
cosine.sim <- function(u,v) {  
  numerator <- u %*% v  
  denominator <- sqrt(u %*% u) * sqrt(v %*% v)  
  return (numerator/denominator)  
}
```

```
cosine.sim(u,v)
```

```
##           [,1]
```

```
## [1,] -0.2520848
```

Cosine similarity

Cosine similarity between Marx and Durkheim

We can use the two rows of the DTM matrix defined above as arguments to the similarity function.

```
print(cosine.sim(Xm[1,], Xm[2,]))
```

```
##           [,1]
```

```
## [1,] 0.1288304
```


Cosine similarity

Cosine similarity for a larger corpus

Let's consider another example with a larger corpus of texts.

```
m <- gutenbergs_metadata |>
  filter(author == "Shakespeare, William" & language == "en")
plays <- gutenbergs_download(2235:2269)

plays <- plays |> left_join(m, by = "gutenberg_id") |>
  filter(gutenberg_id != 2240) # Removing a duplicate
```

Cosine similarity

From text to DTM

Exercise: Modify the pipeline to filter out words that occur less than 5 times across the corpus. Inspect the object first.

Cosine similarity

From text to DTM

Inspect `plays.tidy`. Which words get the highest IDF scores?

Cosine similarity

Creating the term-frequency matrix

```
DTM <- plays.tidy |> cast_dtm(title, word, tf)
DTMd <- as.matrix(DTM)
```

Cosine similarity

Normalizing document vectors

We can simplify the cosine similarity calculation if we normalize each document vector by its length (the denominator in the above calculation.)

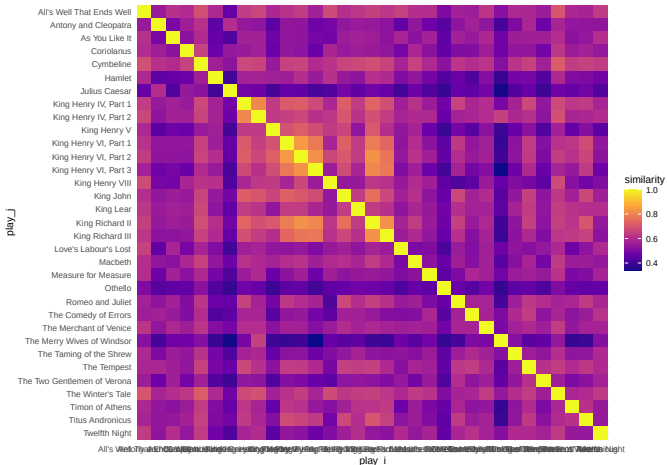
```
normalize <- function(v) {  
  return (v/sqrt(v %*% v))  
}  
  
# Normalizing every row (document) in the matrix  
for (i in 1:dim(DTMd)[1]) {  
  DTMd[i,] <- normalize(DTMd[i,])  
}
```

Cosine similarity

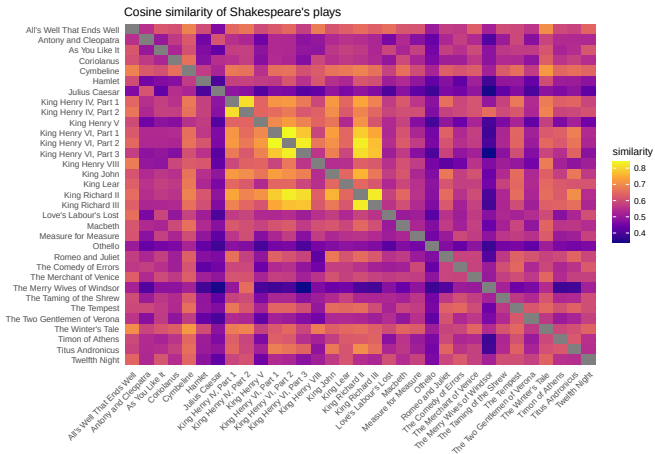
Calculating cosine similarity using matrix multiplication

```
sims <- DTMd %*% t(DTMd)
```

Cosine similarity



Cosine similarity



Next week

- ▶ Word embeddings and computational semantics